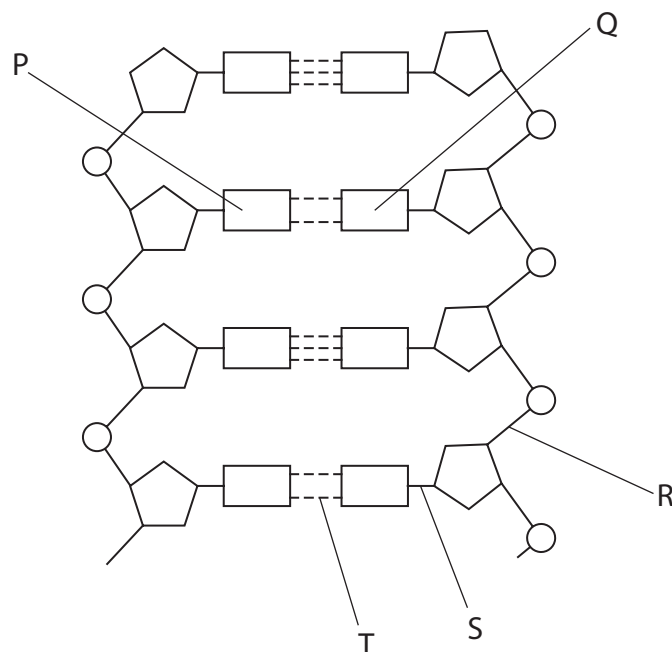


2 Errors in DNA replication can give rise to mutations.

(a) The diagram shows part of a DNA molecule.



(i) The base labelled **Q** is thymine.

Which is the base labelled **P**?

(1)

- ☐ **A** adenine
- ☐ **B** cytosine
- ☐ **C** guanine
- ☐ **D** uracil

(ii) Which row of the table identifies the bonds labelled **R**, **S** and **T**?

(1)

	R	S	T
☐ A	covalent	phosphodiester	hydrogen
☐ B	hydrogen	covalent	phosphodiester
☐ C	hydrogen	phosphodiester	covalent
☐ D	phosphodiester	covalent	hydrogen



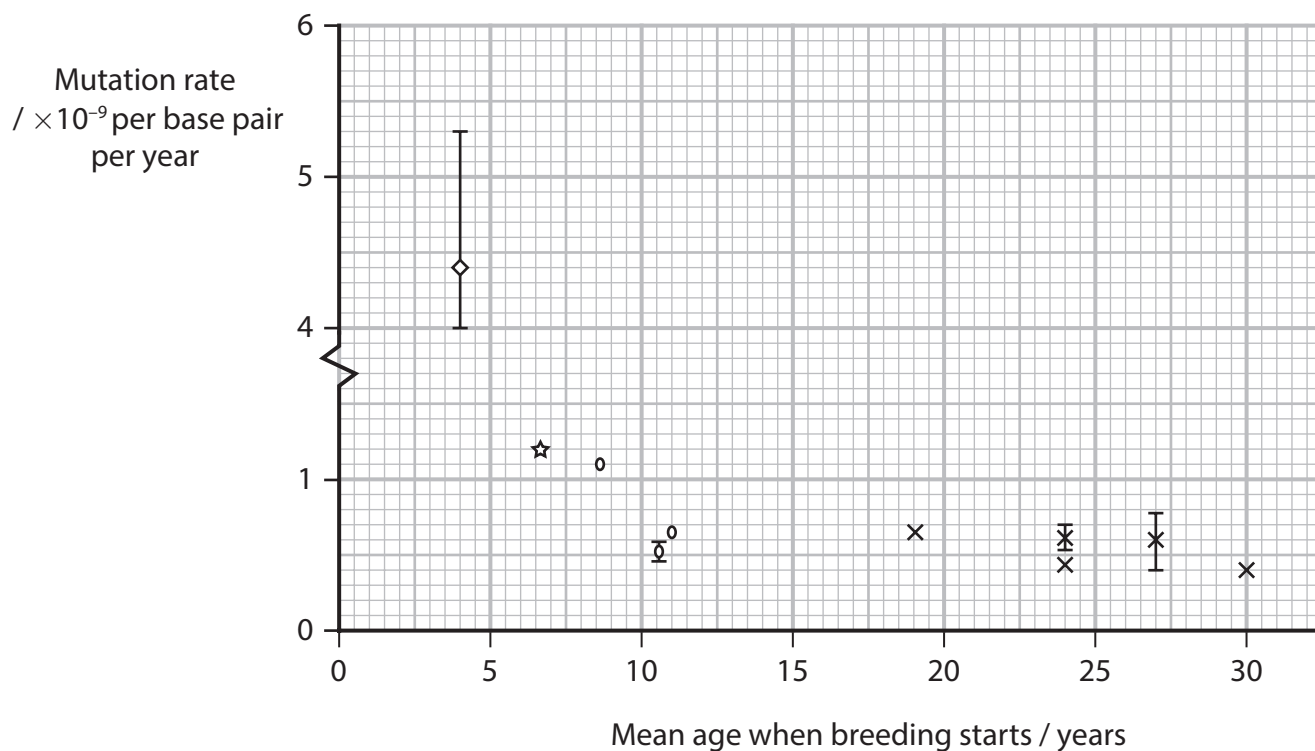
(b) Name **two** types of gene mutation.

(1)

1

2

(c) The graph shows the estimated mutation rate in four groups of primates.



Key

◇ prosimians

☆ new world monkeys

○ old world monkeys

× apes

Describe **two** conclusions that can be made about the mutation rate in these four groups of primates.

(2)

1

.....

.....

2

.....

.....

(Total for Question 2 = 5 marks)



P 7 5 7 7 1 A 0 5 2 8